

1 Physical quantities and units

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A physical quantity combines a number and a unit. Base SI quantities include mass, length, time and current; derived units are products of base units. Dimensional homogeneity checks equations, while percentage uncertainties add for products or quotients. Precision describes repeatability; accuracy concerns closeness to the true value.

Worked example: follow the reasoning

For $a = F/m$, $\text{N kg}^{-1} = (\text{kg m s}^{-2})/\text{kg} = \text{m s}^{-2}$. If F has 3% and m has 2% uncertainty, a has about 5% percentage uncertainty.

Quick check · multiple choice

A Which quantity has SI base unit kg ?

[1]

- A Mass
- B Force
- C Pressure
- D Energy

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A resistance is calculated from $6.0 \pm 0.2 \text{ V}$ and $2.0 \pm 0.1 \text{ A}$. Determine R and its approximate absolute uncertainty.

[3]

Final answer: _____

Answers and explanation

Quick check: A. Mass

Question 1 [3 marks]

$R = V/I = 3.0 \, \Omega$ [1]. Percentage uncertainty = $0.2/6.0 + 0.1/2.0 = 0.0833$ [1]. Absolute uncertainty $\approx 0.25 \, \Omega$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.